# LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

## Generate a Research Paper with AI, Then Audit Its Citations

*AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40*

| Student Detail | Fill in |
|---|---|
| Name | Pranav kumar Gupta |
| Roll Number | MHM2026012 |
| Programme / Department | Human machine interaction |
| Topic ID | 2 |
| Full Assigned Topic | Context-Window Degradation and Its Impact on Long-Document Research Synthesis |
| AI Tool Used | Google Gemini |
| Date | 21_aug_2026 |

**IMPORTANT:** You will receive a topic from T1-T40. Your task is not to produce the "best" paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

## 1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

**Question 1 - Citation Authenticity:** Does the cited publication actually exist, and is it correctly identified?

**Question 2 - Claim-Citation Support:** If the publication is genuine, does it actually support the claim for which the AI cited it?

## 2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

**Most important principle:** You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

## 3. WHAT YOU WILL DO

| Part | Activity |
|---|---|
| A | Record your baseline perception of AI citations |
| B | Record the AI environment used |

| | |
|---|---|
| C | Generate and preserve an AI-written research paper |
| D | Create a reference inventory |
| E | Select 10 references systematically |
| F | Judge their plausibility before verification |
| G | Verify authenticity, metadata and identifiers |
| H | Determine whether genuine references support the AI-generated claims |

## 4. RECOMMENDED 2-HOUR TIME PLAN

| Time | Activity |
|---|---|
| 0-10 min | Parts A-B: Baseline and AI environment |
| 10-30 min | Part C: Generate and save research paper |
| 30-40 min | Parts D-E: Inventory and select references |
| 40-45 min | Part F: Pre-verification plausibility |
| 45-80 min | Part G: Authenticity and metadata audit |
| 80-110 min | Part H: Claim-citation entailment audit |
| 110-120 min | Calculate scores, check worksheet and prepare submission |

**Do not spend excessive time formatting the AI-generated paper.** The audit is the main activity.

## PART A - BASELINE ASSESSMENT

**Complete this section BEFORE opening the AI tool.**

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ✔ Weekly ☐ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ✔ Occasionally ☐ Once or twice ☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ✔ Rarely ☐ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☐ 3 ✔ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

**Important:** Do not change this answer later, even if your opinion changes during the experiment.

## PART B - RECORD YOUR AI ENVIRONMENT

| Information | Your Response |
|---|---|
| AI tool used | Google gemini |
| Model name | 3.1 Pro |
| Model/version shown | 3.1 Pro |
| Access tier | Free |
| Web browsing available? | Yes |
| Was web browsing actually used? | Yes |
| Research/Deep Research mode used? | Yes |
| Date/time generated | 21 Aug 2026 / 10:00AM |

**Important:** If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

# PART C - GENERATE THE AI RESEARCH PAPER

## C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

## C2. Use the Following Common Prompt

**Replace only [YOUR ASSIGNED TOPIC].**

> *Generate a scholarly research paper on the topic: "[YOUR ASSIGNED TOPIC]". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.*

## C3. Preserve the Original Output

- ✔ Save the complete AI-generated paper
- ✔ Save the complete reference list
- ✔ Take screenshot(s) showing the generated output/references
- ✔ Save conversation/export/share evidence if available

**STOP:** Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

# PART D - PAPER PROFILE AND REFERENCE INVENTORY

**Paper title:** Context-Window Degradation and Its Impact on Long-Document Research Synthesis

| Measure | Value |
|---|---|
| Approximate pages | 5 |
| Approximate word count | 1400 |
| Number of sections | 12 |
| Total number of references | 6 |
| Approximate in-text citation occurrences | Not found |
| Did AI warn you to verify references? | No |

## D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

| Reference Type | Count |
|---|---|
| DOI available | 6 |
| PMID available, no DOI | 0 |
| arXiv ID available, no DOI/PMID | 0 |
| URL only | 0 |
| No persistent identifier supplied | 0 |
| TOTAL | 6 |

**Examples**

| Category | Example | How to Count |
|---|---|---|

| | | |
|---|---|---|
| DOI available | Kather, J. N., et al. (2019). Predicting survival from colorectal cancer histology slides using deep learning. PLoS Medicine. DOI: 10.1371/journal.pmed.1002730 | DOI available |
| PMID available, no DOI | A biomedical paper for which the AI provides PMID: 12345678, but no DOI is supplied/found | PMID available, no DOI |
| arXiv ID available | Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629 | arXiv ID available |
| URL only | World Health Organization. Colorectal cancer. Available at: https://... with no DOI, PMID or arXiv ID | URL only |
| No persistent identifier | Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press. | No persistent identifier if none is supplied |

**Important:** Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

## PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

**If the bibliography contains 10 or fewer references:** audit all references.

**If it contains more than 10 references:** select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 8, 13, 18, 23, 26, 27, 28.

| Audit Slot | Original Bibliography Reference No. |
|---|---|
| R1 | 2411.19360 |
| R2 | 10.1162/tacl_a_00638/119630 |
| R3 | 2502.05167 |
| R4 | 2509.21361 |
| R5 | 2506.08479 |
| R6 | 2410.18050 |
| R7 | |
| R8 | |
| R9 | |
| R10 | |

**Important:** From this point onward, R1-R10 refers to your audit sample, not necessarily bibliography references 1-10.

## PART F - PLAUSIBILITY TEST

**Do this BEFORE searching for any reference.**

Look only at the citation as generated by the AI. Ask yourself: “If I had not verified this citation, how convincing would it look to me?”

| Score | Meaning |
|---|---|
| 1 | Obviously suspicious |
| 2 | Somewhat suspicious |
| 3 | Unsure |
| 4 | Looks genuine |
| 5 | Completely convincing |

| Ref. | Initial Impression | Plausibility 1-5 | Predict Genuine? |
|---|---|---|---|
| R1 | | 4 | Yes |
| R2 | | 5 | Yes |

| | | | |
|---|---|---|---|
| R3 | | 3 | Yes |
| R4 | | 5 | Yes |
| R5 | | 4 | Yes |
| R6 | | 5 | Yes |
| R7 | | | |
| R8 | | | |
| R9 | | | |
| R10 | | | |

**Example:** A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

**Critical rule:** Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

## PART G - CITATION AUTHENTICITY AND METADATA AUDIT

### G1. What Are You Checking?

- Does the publication exist?
- Is the title correct?
- Are the authors correct?
- Is the year correct?
- Is the journal/conference/source correct?
- Are volume, issue and pages correct, if supplied?
- Does the DOI/arXiv/PMID belong to this publication?

### G2. Verification Procedure

☐ Persistent identifier: DOI / arXiv / PMID

☐ Publisher or official repository

☐ Google Scholar

☐ Semantic Scholar / OpenAlex

☐ Crossref

☐ PubMed or another discipline-specific database

☐ Exact-title search

☐ Author + title-keyword search

**Not acceptable as sole evidence**: “I asked another AI and it said the paper exists.” AI can help you search, but final verification must rely on scholarly or publisher evidence.

### G3. Assign One A-E Code

| Code | Classification | Meaning |
|---|---|---|
| A | VERIFIED | Publication exists and important metadata are correct. |
| B | WRONG METADATA | Publication exists, but one or more bibliographic details are incorrect. |
| C | FRANKENSTEIN | Real-looking pieces are combined, but that specific publication does not exist. |
| D | FABRICATED | No reliable scholarly evidence that the publication exists. |

| | | |
|---|---|---|
| E | IDENTIFIER MISMATCH | DOI/arXiv/PMID is invalid or resolves to a different publication. |

## G4. Examples

**A:** AI gives a paper and all details match the publisher record.

**B:** The paper exists, but the AI gives the wrong publication year or journal.

**C:** The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

**D:** You cannot find the paper or a credible scholarly record after systematic searching.

**E:** The DOI supplied by AI opens an entirely different paper or does not resolve.

## G5. Record Your Evidence

| Ref. | Publication Found? | Title Match? | Authors Match? | Year Match? | Venue Match? | Identifier Match? |
|---|---|---|---|---|---|---|
| R1 | Y | Y | Y | Y | Y | Y |
| R2 | Y | Y | Y | Y | Y | Y |
| R3 | Y | Y | Y | Y | Y | Y |
| R4 | Y | Y | Y | Y | Y | Y |
| R5 | Y | Y | Y | Y | Y | Y |
| R6 | Y | Y | Y | Y | Y | Y |
| R7 | | | | | | |
| R8 | | | | | | |
| R9 | | | | | | |
| R10 | | | | | | |

| Ref. | Verification Source(s) | Identifier Checked | What You Found |
|---|---|---|---|
| R1 | The paper **“DENIAHL: In-Context Features Influence LLM Needle-In-A-Haystack Abilities”** exists as arXiv:2411.19360. The authors and year in your paper match the arXiv record. | VERIFIED | Publication exists and the supplied bibliographic information and identifier match. |
| R2 | The paper **“NoLiMa: Long-Context Evaluation Beyond Literal Matching”** exists as arXiv:2502.05167.<br><br>The authors, title, year and identifier match. The paper's actual findings also match the claim in your research paper: performance drops substantially as context length increases, and at 32K tokens many models fall below 50% of their short-context baseline. | VERIFIED | Publication exists, metadata/identifier match, and the cited claim is supported. |
| R3 | The paper **“NoLiMa: Long-Context Evaluation** | VERIFIED | Publication exists, metadata/identifier match, and the cited claim is supported. |

| | | | |
|---|---|---|---|
| | **Beyond Literal Matching"** exists as arXiv:2502.05167.<br><br>The authors, title, year and identifier match. The paper's actual findings also match the claim in your research paper: performance drops substantially as context length increases, and at 32K tokens many models fall below 50% of their short-context baseline. | | |
| R4 | The paper **"Context Is What You Need: The Maximum Effective Context Window for Real World Limits of LLMs"** exists as arXiv:2509.21361.<br><br>The author **Norman Paulsen**, year **2025**, title and arXiv identifier all match. The paper does report a significant difference between maximum context window and maximum effective context window, including severe degradation around 1,000 tokens for many tested models.<br><br>**Final code: A — VERIFIED** | VERIFIED | Publication exists and the supplied metadata/identifier match. |
| R5 | The paper **"Efficient Context Selection for Long-Context QA: No Tuning, No Iteration, Just Adaptive-k"** exists as arXiv:2506.08479.<br><br>The authors are:<br><br>● Chihiro Taguchi<br>● Seiji Maekawa<br>● Nikita Bhutani<br><br>The year and identifier match your reference. Its actual research describes Adaptive-k as dynamically selecting the number of passages based on similarity-score distributions and reducing token usage | VERIFIED | Publication exists, metadata/identifier match, and the cited claim is supported. |

| | | | |
|---|---|---|---|
| | while maintaining relevant retrieval. | | |
| R6 | The paper **"LongRAG: A Dual-Perspective Retrieval-Augmented Generation Paradigm for Long-Context Question Answering"** exists as arXiv:2410.18050.<br><br>The authors in your PDF match the verified record:<br><br>Qingfei Zhao, Ruobing Wang, Yukuo Cen, Daren Zha, Shicheng Tan, Yuxiao Dong, and Jie Tang.<br><br>The paper specifically discusses the problem of ordinary RAG chunking disrupting global information and proposes a dual-perspective approach addressing global information and factual details. | VERIFIED | Publication exists, metadata/identifier match, and the cited claim is supported. |
| R7 | | | |
| R8 | | | |
| R9 | | | |
| R10 | | | |

## G6. Final Classification

| Ref. | Final Code | One-Line Reason |
|---|---|---|
| R1 | A | Publication exists and the title, authors, year and arXiv identifier match. |
| R2 | A | Publication exists and all important bibliographic details and DOI match. |
| R3 | A | Publication exists and the metadata/identifier match; cited findings are supported. |
| R4 | A | Publication exists and the title, author, year and arXiv identifier match. |
| R5 | A | Publication exists and the metadata/identifier match; Adaptive-k claim is supported. |
| R6 | A | Publication exists and the metadata/identifier match; LongRAG claim is supported. |
| R7 | A/B/C/D/E | |

| | | |
|---|---|---|
| R8 | A/B/C/D/E | |
| R9 | A/B/C/D/E | |
| R10 | A/B/C/D/E | |

**Good reason:** B: Article exists, but AI reported 2022 instead of 2020.

**Poor reason:** B: Details wrong.

## PART G - SCORING

Count your final classifications: A = 6 B =0 C = 0 D = 0 E =0

Total audited: N = A + B + C + D + E = __6__

| Code | Points |
|---|---|
| A | 6 |
| B | 0 |
| C | 0 |
| D | 0 |
| E | 0 |

Authenticity Points = 4A + 3B + C + E

Authenticity Score = [(4A + 3B + C + E) / (4N)] x 100

My calculation: 4(_6___) + 3(___0_) + 1(__0__) + 0(___0_) + 1(__0__) = _24___ points

Authenticity Score = __24__ / (4 x __6__) x 100 = ____100__ /100

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = 4(6)+3(2)+1(1)=31; maximum = 40; Authenticity Score = 31/40 x 100 = 77.5.

| Score | Interpretation |
|---|---|
| 90-100 | Excellent authenticity |
| 75-89 | Good, minor concerns |
| 60-74 | Moderate citation-authenticity risk |
| 40-59 | High citation-authenticity risk |
| Below 40 | Very high citation-authenticity risk |

### G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = (Correct genuine/fake predictions / Total predictions) x 100

Correct predictions = _6___ Total predictions = __6__ Prediction Accuracy = ___100___ %

# FINAL LAB 1 RESULTS

| Measure | Your Result |
|---|---|
| Total references in AI paper | 6 |
| References deeply audited | 6 |
| A: Verified | 6 |
| B: Wrong metadata | 0 |
| C: Frankenstein | 0 |
| D: Fabricated | 0 |
| E: Identifier mismatch | 0 |

| | |
|---|---|
| Authenticity Score | 100 /100 |
| Prediction Accuracy | 100 % |

## FINAL INTERPRETATION

**1. Did professional-looking citations necessarily turn out to be genuine?**

In this particular sample, **yes—all six professional-looking citations turned out to be genuine**. However, this audit alone does not prove that AI-generated citations are always reliable.

**2. Did all genuine references actually support the claims for which they were cited?**

**Yes.** The six audited references were found to support the major claims associated with them.

**3. What was the most serious citation-integrity problem you observed?**

**No serious citation-integrity problem was found in the six-reference sample.** All six were classified A.

**4. What is the single most important lesson you learned from this lab?**

AI-generated citations should still be independently verified. A citation looking genuine is not sufficient evidence; both the existence of the source and its support for the specific claim must be checked

## SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

## STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

**Student Name:Pranav Kumar Gupta**

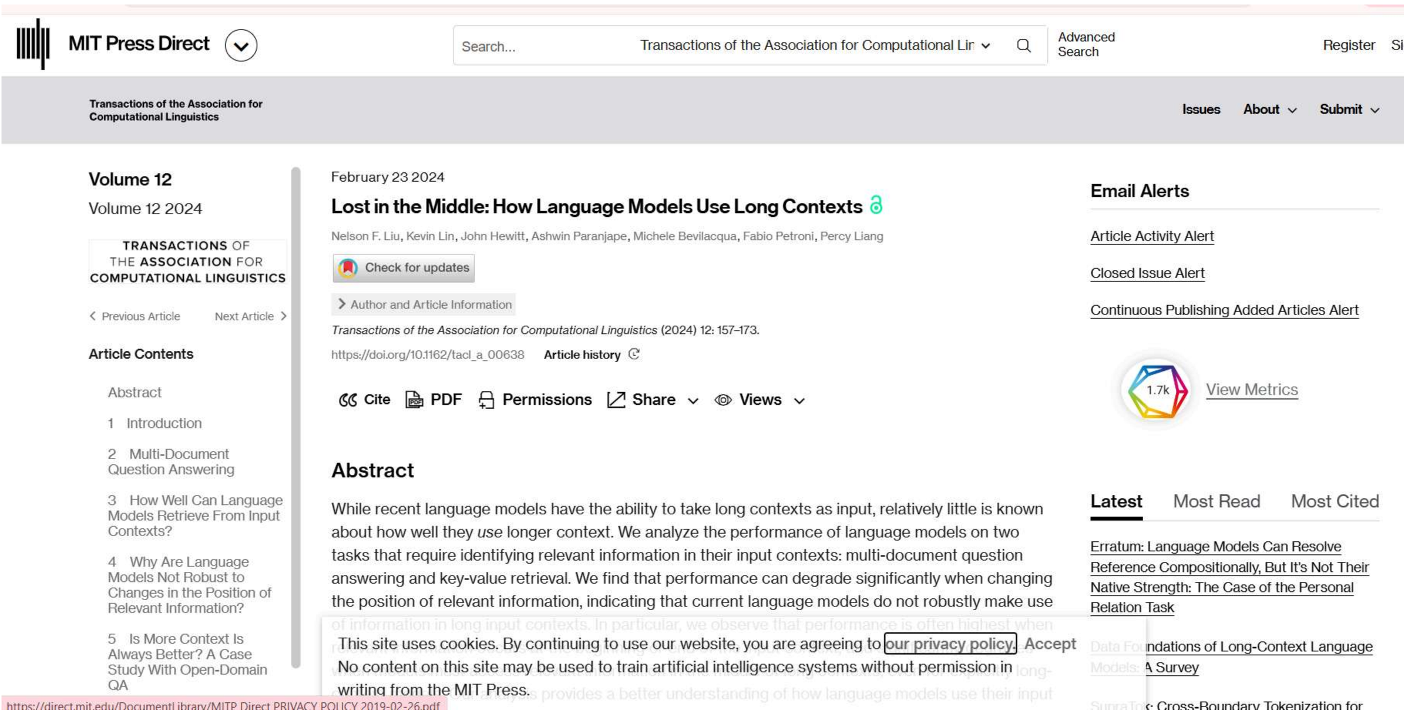
**Roll Number:** MHM2026012

**Signature: .Pranav Kumar Gupta**

**Date:** .21 AUG 2026

# CENTRAL LESSON OF LAB 1

## Does the source exist?



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Article Contents

Abstract

1 Introduction

2 Multi-Document Question Answering

3 How Well Can Language Models Retrieve From Input Contexts?

4 Why Are Language Models Not Robust to Changes in the Position of Relevant Information?

5 Is More Context Is Always Better? A Case Study With Open-Domain QA

February 23 2024

**Lost in the Middle: How Language Models Use Long Contexts**

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, Percy Liang

Check for updates

> Author and Article Information

*Transactions of the Association for Computational Linguistics* (2024) 12: 157–173.

https://doi.org/10.1162/tacl_a_00638 Article history

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**Abstract**

While recent language models have the ability to take long contexts as input, relatively little is known about how well they *use* longer context. We analyze the performance of language models on two tasks that require identifying relevant information in their input contexts: multi-document question answering and key-value retrieval. We find that performance can degrade significantly when changing the position of relevant information, indicating that current language models do not robustly make use of information in long input contexts. In particular, we observe that performance is often highest when

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... provides a better understanding of how language models use their input

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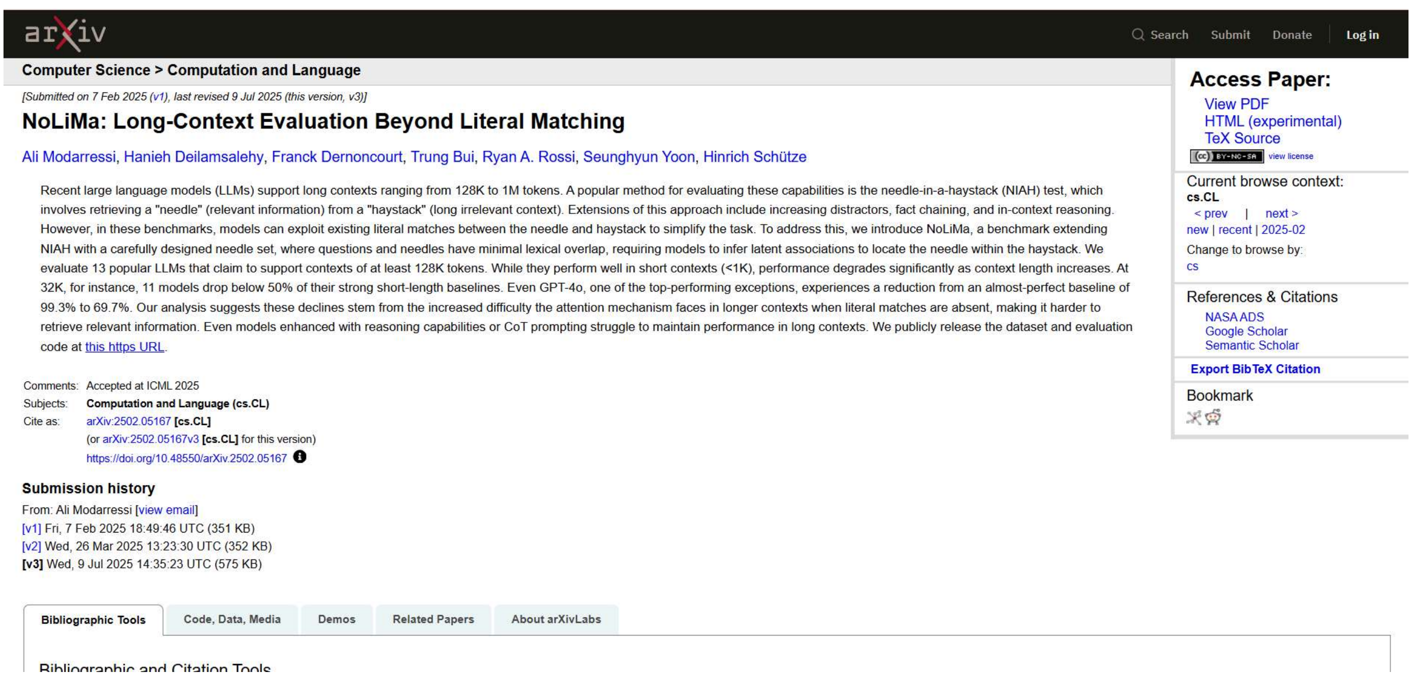
Erratum: Language Models Can Resolve Reference Compositionally, But It's Not Their Native Strength: The Case of the Personal Relation Task

Data Foundations of Long-Context Language Models: A Survey

...: Cross-Boundary Tokenization for

https://direct.mit.edu/DocumentLibrary/MITP Direct PRIVACY POLICY 2019-02-26.pdf



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**Computer Science > Computation and Language**

*[Submitted on 7 Feb 2025 (v1), last revised 9 Jul 2025 (this version, v3)]*

**NoLiMa: Long-Context Evaluation Beyond Literal Matching**

Ali Modarressi, Hanieh Deilamsalehy, Franck Dernoncourt, Trung Bui, Ryan A. Rossi, Seunghyun Yoon, Hinrich Schütze

Recent large language models (LLMs) support long contexts ranging from 128K to 1M tokens. A popular method for evaluating these capabilities is the needle-in-a-haystack (NIAH) test, which involves retrieving a "needle" (relevant information) from a "haystack" (long irrelevant context). Extensions of this approach include increasing distractors, fact chaining, and in-context reasoning. However, in these benchmarks, models can exploit existing literal matches between the needle and haystack to simplify the task. To address this, we introduce NoLiMa, a benchmark extending NIAH with a carefully designed needle set, where questions and needles have minimal lexical overlap, requiring models to infer latent associations to locate the needle within the haystack. We evaluate 13 popular LLMs that claim to support contexts of at least 128K tokens. While they perform well in short contexts (<1K), performance degrades significantly as context length increases. At 32K, for instance, 11 models drop below 50% of their strong short-length baselines. Even GPT-4o, one of the top-performing exceptions, experiences a reduction from an almost-perfect baseline of 99.3% to 69.7%. Our analysis suggests these declines stem from the increased difficulty the attention mechanism faces in longer contexts when literal matches are absent, making it harder to retrieve relevant information. Even models enhanced with reasoning capabilities or CoT prompting struggle to maintain performance in long contexts. We publicly release the dataset and evaluation code at this https URL.

Comments: Accepted at ICML 2025

Subjects: **Computation and Language (cs.CL)**

Cite as: arXiv:2502.05167 **[cs.CL]**

(or arXiv:2502.05167v3 **[cs.CL]** for this version)

https://doi.org/10.48550/arXiv.2502.05167

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*[Submitted on 28 Nov 2024]*

## DENIAHL: In-Context Features Influence LLM Needle-In-A-Haystack Abilities

Hui Dai, Dan Pechi, Xinyi Yang, Garvit Banga, Raghav Mantri

The Needle-in-a-haystack (NIAH) test is a general task used to assess language models' (LMs') abilities to recall particular information from long input context. This framework however does not provide a means of analyzing what factors, beyond context length, contribute to LMs' abilities or inabilities to separate and recall needles from their haystacks. To provide a systematic means of assessing what features contribute to LMs' NIAH capabilities, we developed a synthetic benchmark called DENIAHL (Data-oriented Evaluation of NIAH for LLM's). Our work expands on previous NIAH studies by ablating NIAH features beyond typical context length including data type, size, and patterns. We find stark differences between GPT-3.5 and LLaMA 2-7B's performance on DENIAHL, and drops in recall performance when features like item size are increased, and to some degree when data type is changed from numbers to letters. This has implications for increasingly large context models, demonstrating factors beyond item-number impact NIAH capabilities.

Subjects: **Computation and Language (cs.CL)**; Artificial Intelligence (cs.AI); Machine Learning (cs.LG)
Cite as: arXiv:2411.19360 **[cs.CL]**
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Computer Science > Computation and Language

*[Submitted on 21 Sep 2025 (v1), last revised 23 Apr 2026 (this version, v2)]*

## Context Is What You Need: The Maximum Effective Context Window for Real World Limits of LLMs

Norman Paulsen

Large language model (LLM) providers boast big numbers for maximum context window sizes. To test the real world use of context windows, we 1) define a concept of maximum effective context window, 2) formulate a testing method of a context window's effectiveness over various sizes and problem types, and 3) create a standardized way to compare model efficacy for increasingly larger context window sizes to find the point of failure. We collected hundreds of thousands of data points across several models and found significant differences between reported Maximum Context Window (MCW) size and Maximum Effective Context Window (MECW) size. Our findings show that the MECW is, not only, drastically different from the MCW but also shifts based on the problem type. A few top of the line models in our test group failed with as little as 100 tokens in context; most had severe degradation in accuracy by 1000 tokens in context. All models fell far short of their Maximum Context Window by as much as 99 percent. Our data reveals the Maximum Effective Context Window shifts based on the type of problem provided, offering clear and actionable insights into how to improve model accuracy and decrease model hallucination rates.

Comments: 20 pages, 4 charts. AAIML (2026)
Subjects: **Computation and Language (cs.CL)**; Artificial Intelligence (cs.AI)
ACM classes: I.2.7
Cite as: arXiv:2509.21361 **[cs.CL]**
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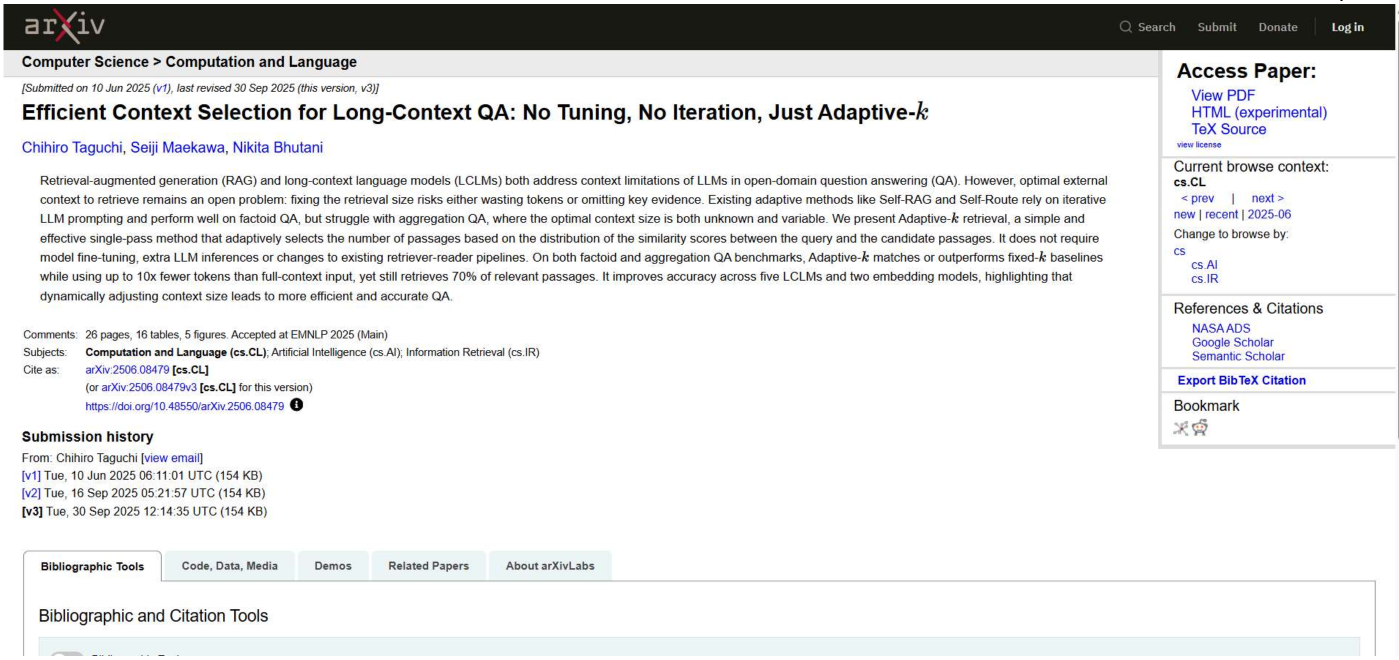

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[Submitted on 10 Jun 2025 (v1), last revised 30 Sep 2025 (this version, v3)]

**Efficient Context Selection for Long-Context QA: No Tuning, No Iteration, Just Adaptive-$k$**

Chihiro Taguchi, Seiji Maekawa, Nikita Bhutani

Retrieval-augmented generation (RAG) and long-context language models (LCLMs) both address context limitations of LLMs in open-domain question answering (QA). However, optimal external context to retrieve remains an open problem: fixing the retrieval size risks either wasting tokens or omitting key evidence. Existing adaptive methods like Self-RAG and Self-Route rely on iterative LLM prompting and perform well on factoid QA, but struggle with aggregation QA, where the optimal context size is both unknown and variable. We present Adaptive-$k$ retrieval, a simple and effective single-pass method that adaptively selects the number of passages based on the distribution of the similarity scores between the query and the candidate passages. It does not require model fine-tuning, extra LLM inferences or changes to existing retriever-reader pipelines. On both factoid and aggregation QA benchmarks, Adaptive-$k$ matches or outperforms fixed-$k$ baselines while using up to 10x fewer tokens than full-context input, yet still retrieves 70% of relevant passages. It improves accuracy across five LCLMs and two embedding models, highlighting that dynamically adjusting context size leads to more efficient and accurate QA.

Comments: 26 pages, 16 tables, 5 figures. Accepted at EMNLP 2025 (Main)
Subjects: **Computation and Language (cs.CL)**; Artificial Intelligence (cs.AI); Information Retrieval (cs.IR)
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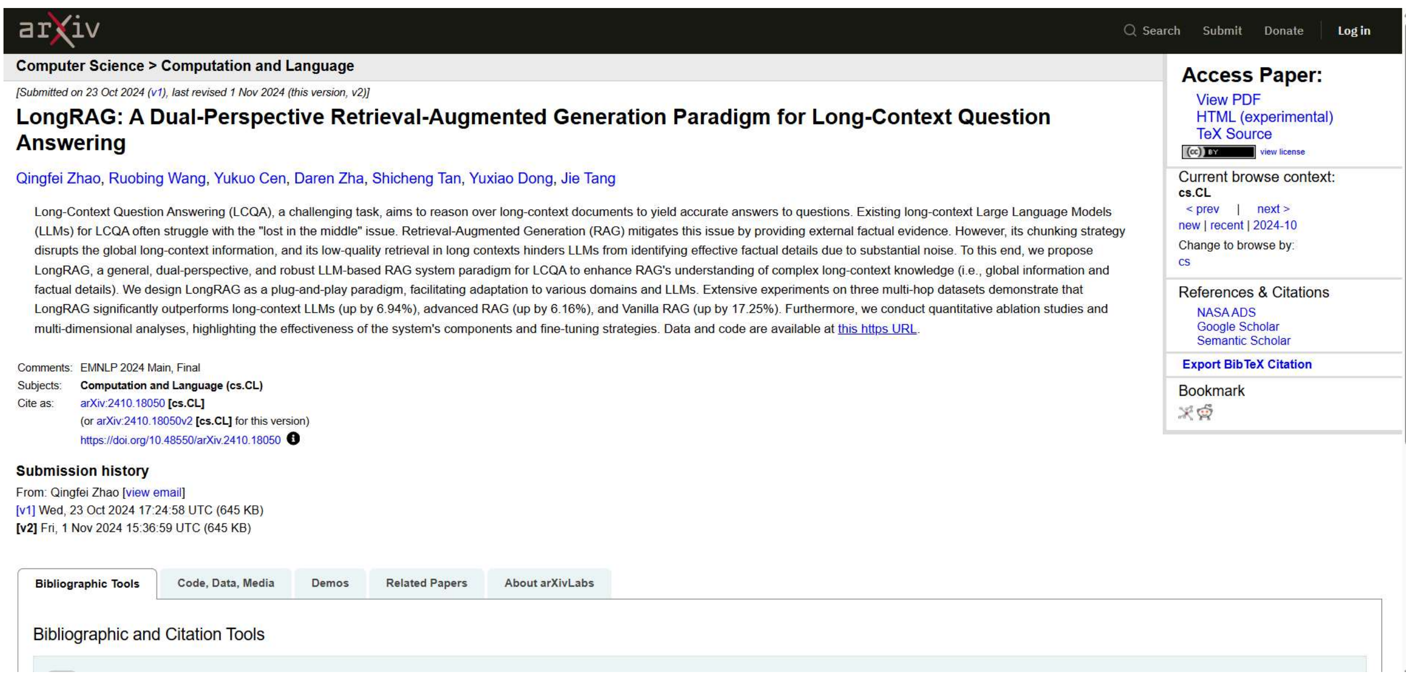

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[Submitted on 23 Oct 2024 (v1), last revised 1 Nov 2024 (this version, v2)]

**LongRAG: A Dual-Perspective Retrieval-Augmented Generation Paradigm for Long-Context Question Answering**

Qingfei Zhao, Ruobing Wang, Yukuo Cen, Daren Zha, Shicheng Tan, Yuxiao Dong, Jie Tang

Long-Context Question Answering (LCQA), a challenging task, aims to reason over long-context documents to yield accurate answers to questions. Existing long-context Large Language Models (LLMs) for LCQA often struggle with the "lost in the middle" issue. Retrieval-Augmented Generation (RAG) mitigates this issue by providing external factual evidence. However, its chunking strategy disrupts the global long-context information, and its low-quality retrieval in long contexts hinders LLMs from identifying effective factual details due to substantial noise. To this end, we propose LongRAG, a general, dual-perspective, and robust LLM-based RAG system paradigm for LCQA to enhance RAG's understanding of complex long-context knowledge (i.e., global information and factual details). We design LongRAG as a plug-and-play paradigm, facilitating adaptation to various domains and LLMs. Extensive experiments on three multi-hop datasets demonstrate that LongRAG significantly outperforms long-context LLMs (up by 6.94%), advanced RAG (up by 6.16%), and Vanilla RAG (up by 17.25%). Furthermore, we conduct quantitative ablation studies and multi-dimensional analyses, highlighting the effectiveness of the system's components and fine-tuning strategies. Data and code are available at this https URL.

Comments: EMNLP 2024 Main, Final
Subjects: **Computation and Language (cs.CL)**
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From: Qingfei Zhao [view email]
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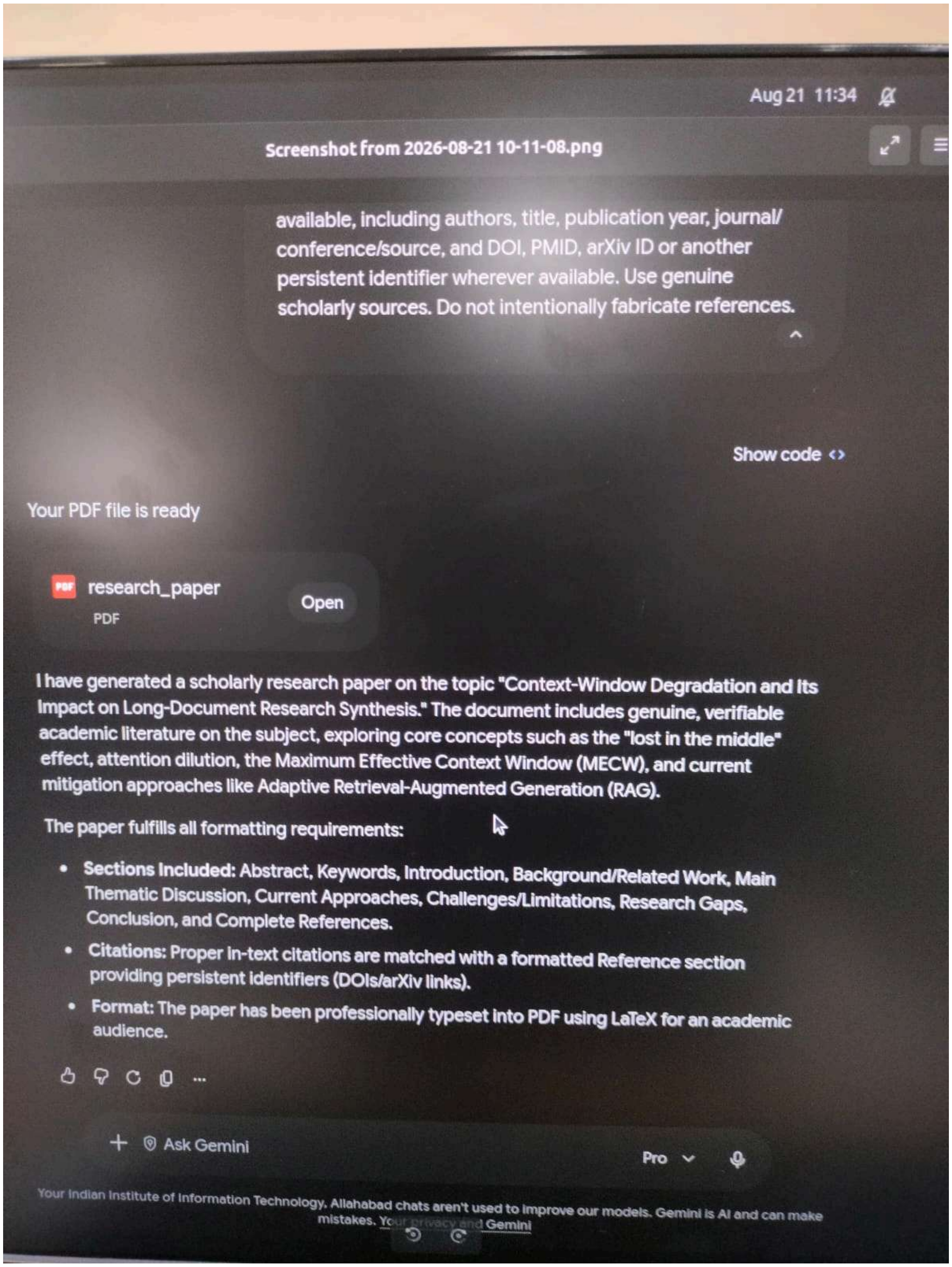


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